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VISION & MISSION OF THE UNIVERSITY

Vision

“To be a world class university through education, innovation and research for the service of humanity.”

Mission

1. To establish Centres of excellence in emerging areas of science, engineering, technology, management and allied areas.
2. To foster an ecosystem for incubation, product development, transfer of entrepreneurship, technology and
3. To create environment of collaboration, experimentation, imagination and creativity.
4. To develop human potential with analytical abilities, ethics and integrity.
5. To provide environment friendly, reasonable and sustainable solutions for local and global needs..

VISION & MISSION OF CSE DEPARTMENT

Vision

Department of Computer Science & Engineering to be a leading world class technology department playing its role as a key node in national and global knowledge network, thus empowering the computer science industry with the wings of knowledge and power of innovation.

Mission

1. To nurture talent of students for research, innovation and excellence in the field of computer engineering starting from Under Graduate level.
2. To develop highly analytical and qualified computer engineers by imparting training on cutting edge technology.
3. To produce socially sensitive computer engineers with professional ethics.
4. To focus on R&D environment in close partnership with industry and foreign universities.
5. To produce well-rounded, up to date, scientifically tempered, design oriented engineers and scientists capable of lifelong learning.

Experiment No. 1

AIM

Study and Overview of Operating System. Discuss functions of Operating System along with different types of Operating System.

THEORY

Introduction to Operating System

An Operating System (OS) is system software that acts as an interface between the computer hardware and the end user. It manages hardware resources such as the CPU, memory, storage devices and input/output devices, and provides a platform on which application programs can run. Without an operating system, a user would have to directly communicate with the hardware using machine-level instructions, which is impractical. The OS hides the complexity of the hardware and provides a convenient, efficient and secure environment for program execution.

Functions of Operating System

The major functions performed by an Operating System are described below:

1. **Process Management:** The OS is responsible for creating, scheduling, suspending, resuming and terminating processes. It allocates CPU time to processes using scheduling algorithms and handles process synchronization and inter-process communication.
2. **Memory Management:** The OS keeps track of which parts of memory are in use and by which process. It allocates and deallocates memory space as needed, and implements techniques such as paging, segmentation and virtual memory to use memory efficiently.
3. **File Management:** The OS manages the creation, deletion, reading, writing and organization of files and directories on secondary storage devices. It also handles file permissions and access control.
4. **Device Management:** The OS manages I/O devices such as keyboards, printers, disk drives and network adapters through device drivers, and ensures efficient and conflict-free access to these devices by multiple processes.
5. **Security and Access Control:** The OS protects the system and user data from unauthorized access through authentication mechanisms, user permissions and access control lists.
6. **Resource Allocation:** When multiple processes require the same resource simultaneously, the OS decides how resources are allocated among them to ensure fairness and efficiency.
7. **User Interface:** The OS provides an interface, either Command Line Interface (CLI) or Graphical User Interface (GUI), through which users can interact with the computer system.
8. **Error Detection and Handling:** The OS continuously monitors the system for errors and handles them so that they do not cause the entire system to fail.

Types of Operating System

1. Batch Operating System: Jobs with similar requirements are grouped together and executed as a batch without user interaction during execution. It reduces idle CPU time but lacks interactivity. Example: early mainframe systems.
2. Time-Sharing (Multitasking) Operating System: The CPU time is divided among multiple users/processes in small time slices (time quantum), giving each user the illusion of having a dedicated system. Example: UNIX.
3. Distributed Operating System: Manages a group of independent computers connected over a network, making them appear to users as a single coherent system, allowing resource sharing across machines. Example: Locus OS.
4. Network Operating System: Runs on a server and provides the capability to manage data, users, groups, security and other networking functions over a network. Examples: Windows Server, Novell NetWare.
5. Real-Time Operating System (RTOS): Designed for applications with strict timing constraints, where tasks must be completed within a specified deadline. Divided into Hard RTOS and Soft RTOS. Examples: VxWorks, RTLinux (used in medical systems, industrial control, avionics).
6. Multiprogramming Operating System: Keeps multiple jobs in main memory simultaneously so that the CPU always has a job to execute, maximizing CPU utilization by switching among jobs when one is waiting for I/O.
7. Multiprocessing Operating System: Uses two or more CPUs within a single computer system to execute multiple processes simultaneously, improving throughput and reliability.
8. Mobile Operating System: Designed specifically for smartphones, tablets and other handheld devices, optimized for touch interaction and limited resources. Examples: Android, iOS.

CODE/QUERY

Not Applicable. This is a theoretical/conceptual experiment based on the study and overview of Operating System concepts; hence no code or query is involved.

OUTPUT

Not Applicable. As this experiment is theoretical in nature, there is no program output. (Insert a screenshot with your name here only if your instructor requires a demonstration screenshot for this experiment.)

VIVA QUESTIONS WITH ANSWERS

Q: What is an Operating System?

Ans: An Operating System is system software that acts as an interface between the user and the computer hardware. It manages hardware resources and provides services to application programs so that the user does not need to interact directly with the hardware.

Q: What are the main functions of an Operating System?

Ans: The main functions include process management, memory management, file management, device management, security and access control, resource allocation, providing a user interface, and error detection and handling.

Q: Differentiate between multiprogramming and multitasking.

Ans: Multiprogramming keeps multiple jobs in memory so the CPU can switch to another job when the current one is waiting for I/O, mainly to maximize CPU utilization. Multitasking (time-sharing) divides CPU time among multiple processes using small time slices so that each user/process appears to run simultaneously, focusing on responsiveness to the user.

Q: What is a Real-Time Operating System (RTOS)? Give examples.

Ans: An RTOS is an operating system designed to process data and respond to inputs within a strict, predictable time limit. It is classified into Hard RTOS (deadlines must never be missed) and Soft RTOS (occasional deadline misses are tolerable). Examples include VxWorks and RTLinux, commonly used in avionics, medical devices and industrial control systems.

Q: What is the difference between a Process and a Program?

Ans: A program is a passive entity - a set of instructions stored on disk, such as an executable file. A process is an active entity - a program in execution, which has its own process control block, CPU registers, memory allocation and current state (new, ready, running, waiting or terminated).